Problem Statement # 1 :

You are tasked with preparing a straight-through Ethernet cable that can be used to connect two different types of networking devices (such as a computer to a switch). The cable must be constructed following the **TIA/EIA-568 standard (either T568A or T568B on both ends)**.

As a networking technician, you need to ensure that:

- The correct color coding sequence is followed on both ends of the cable.
- The cable is properly crimped using appropriate tools.
- The final cable is tested for continuity and correct pin configuration.

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Problem Statement # 2:

A network administrator needs to segment a local area network into multiple VLANs to improve security and performance. However, devices in different VLANs must still communicate with each other. Your task is to design and implement a solution that enables communication between VLANs using a router.

You are required to configure VLANs on multiple (min 2) switches and implement inter-VLAN routing using a router.

Hints:

- The organization has three departments:
  - VLAN 10 – HR (192.168.10.0/24)
  - VLAN 20 – IT (192.168.20.0/24)
  - VLAN 30 – Sales (192.168.30.0/24)
- Create and assign VLANs on the switch.
- Assign appropriate IP addresses to devices in each VLAN.
- Configure trunking between switch .
- Enable routing between VLANs.
- Verify connectivity using ping tests between different VLANs.

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Problem Statement # 3:

To design and implement a client-server communication system using socket programming, following a defined simple communication protocol. A client device sends requests to a server device. The server processes the request and sends an appropriate response based on the defined protocol.

Protocol Requirements:

- Each message must include:
  - A command (e.g., TIME, DATE, MSG, EXIT)
  - Optional data (if required)
- The server must respond according to the command:
  - TIME → returns current server time
  - DATE → returns current server date
  - MSG <text> → echoes the message
  - EXIT → terminates the connection

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